ZUNFT

To-Do for you

The things only you can do, as clear step-by-step guides: accounts to open, people to talk to, decisions to make. Each task says why it matters, what it costs, how long it takes and what to send me afterwards.

State: 14 October 2026 | Companion documents: **Zunft_Guide** and **Zunft_Tests**

Overview: what to do, and in what order

Nothing here blocks the testing on your own phones. These tasks decide when Zunft can safely go to **real customers** and when it can **charge money**.

	Task	Needed before	Cost	Time
☐	**1** Tax advisor meeting (questions are ready)	First real invoice	Advisor's fee	1 h meeting, 1-2 weeks to get one
☐	**2** Data protection basics (contracts with the AI/cloud providers, AI data settings)	Any outside tester with real data	Free	2-3 h
☐	**3** Lawyer review (privacy text, terms, working-time questions)	Selling to customers	Lawyer's fee	Weeks
☐	**4** Supabase for real use (paid plan, confirmation link, e-mail sender)	Any outside tester	Monthly	1-2 h
☐	**5** Signing key and Google Play	Installing on testers' phones the proper way	One-time fee	1 h, plus days of waiting for Google
☑	**6** Firebase for push messages (done; one small step left, see 6.4)	-	Free	-
☐	**7** Prices, plans and payments (by hand first, Stripe later; Play policy check)	Charging money	Per transaction	1 day to decide, 1 day to set up
☐	**8** Domain, e-mail addresses, Impressum	Public launch	Small	1-2 h
☐	**9** Cost alarms and key hygiene	Any outside tester	Free	30 min
☐	**10** The pilot with 2-3 real craftsmen	-	-	2-4 weeks
☐	**11** iPhone version (optional)	When you want iOS	Yearly fee	Needs a Mac
☐	**12** Name check and backups of your own	Any time	Free	1 h

SUGGESTED ORDER

Do **9** and **4** first (a day), then **2**, then book **1** and **3** because they take the longest to get an appointment for. While you wait: **5** and **6**. Run the tests from the Test Plan in parallel. Only after 1, 2, 3 and 4 give the app to outside craftsmen (task 10).

1. Tax advisor meeting

Why: the invoice texts and calculations are proposals until an advisor confirms them **Time:** 1 hour

Zunft prints a § 35a notice, splits labour and material, issues cancellations, sells material onward with a markup and now creates e-invoices. A wrong sentence on thousands of invoices is a real risk. The advisor's answers decide whether any wording or calculation changes.

PREPARE (30 MINUTES)

1. In the app, on your test business, create these examples and send them to yourself by e-mail as PDF: (a) an invoice for a **private** customer with labour, a flat fee and material; (b) an invoice for a **company** customer as an **e-invoice**; (c) a **Stornorechnung** of a test invoice; (d) a quote.
2. Export the receipt archive of one month: Belege → Einkaufsbelege → Exportieren (ZIP). Also share the monthly overview: Betrieb → Übersicht → Als Datei teilen (CSV).
3. Print **Zunft_Steuerberater.pdf** (in the docs folder). It lists about 20 questions with a box for the answer under each. Bring it, the printed examples and the ZIP on a USB stick or e-mailed.

AT THE MEETING

4. Start with the questions that can change the product most: **A1/A2** (flat fee as labour, the § 35a sentence), **C1/C2** (Stornorechnung and paid invoices), **E1/E2** (material bought for the customer, input VAT on till receipts), **E3** (keeping receipts, throwing away the paper), **W1** (e-invoice: ZUGFeRD enough or also XRechnung?), **W6** (billing tracked hours).
5. Ask two extra questions that are not on the list: (a) does the advisor want the export in a particular format (for example DATEV)? (b) does the business need a written **Verfahrensdokumentation** (GoBD process description), and may I draft it from the Project Guide?
6. Write the answers in the boxes. If the advisor says something is wrong, do not argue; just note the exact words.

AFTERWARDS

7. Photograph or type the answers and send them to me. I change the texts and calculations accordingly, add tests, and update the three documents.

GOOD TO KNOW

No answer is needed to keep testing. But do not send a real invoice to a real customer from Zunft before this meeting has happened.

THE LEITWEG-ID (ONLY FOR INVOICES TO PUBLIC AUTHORITIES)

You do not apply for it. It is the *buyer's* routing number: a public authority (city, state office, ministry, school administration) tells you its Leitweg-ID in the order, contract or its invoicing instructions (often on the authority's website under "Rechnungsstellung" or "E-Rechnung"). It looks like numbers separated by hyphens. You need it only for the pure-XML **XRechnung** format that public bodies require; private companies are served by the ZUGFeRD e-invoice Zunft already makes. Ask the advisor in question W1 whether you will have public customers at all. If you get your first one, send me the Leitweg-ID format they use and I add the field and the XRechnung export.

2. Data protection basics (you can do this yourself)

Why: customer names, addresses and voices go to outside services **Cost:** free **Time:** 2-3 hours

Where the data goes: the **phone** → **Supabase** (database and files; your project is in the EU, Ireland) → **Deepgram** (voice to text) and **Google Gemini** (understanding text and receipt photos). The craftsmen who use Zunft are the *controllers* of their customers' data; you are their *processor*, and Deepgram, Google and Supabase are your *sub-processors*. You need a signed data-processing agreement (AVV / DPA) with each of the three, and the AI settings must be right.

2.1 SUPABASE

1. Open `supabase.com/dashboard`, click your organisation name (top left) → **Organization settings** → **Legal Documents** (also reachable through the "Legal" or "DPA" entry). Read the DPA and use the button to sign or request it.
2. Save the signed PDF in a folder called `Zunft-Vertraege` on your computer and in cloud storage.
3. Check the region once: project **zunft** → **Settings** → **General**: it must say a EU region (it does: eu-west-1, Ireland). Do not create the production project anywhere else.

2.2 GOOGLE GEMINI (THE MOST IMPORTANT SETTING)

1. Open `aistudio.google.com` → **API keys**. Find the key that Zunft uses and look at which **project** it belongs to.
2. Click **Set up billing** for that project (it opens Google Cloud). As far as I know, Google's terms treat content sent on the **free tier** differently from the **paid tier**: on the free tier Google may use the content to improve its products; on the paid tier it does not. Real customer data must only go through a project with billing enabled. **Please read Google's current "Gemini API Additional Terms" yourself**, because these terms change.
3. In the Google Cloud console for that project: **Billing** → **Budgets & alerts** → create a monthly budget (for example 20 EUR) with e-mail alerts at 50 %, 90 % and 100 %.
4. Find and accept the **Cloud Data Processing Addendum** (Google Cloud Console → **IAM & Admin** → **Privacy & Security** or the "Legal" page of Google Cloud). Save the confirmation.

2.3 DEEPGRAM

1. Open `console.deepgram.com`. Look in **Settings** / **Project** for any option about "model improvement" or data retention. Make sure you are **not** taking part in a programme that lets Deepgram keep or train on your audio. If you cannot find it, write to their support and ask in plain words: "Do you store or use our audio for training? Please send your DPA and your list of sub-processors."
2. Ask for the signed DPA and save it with the others. Ask where the audio is processed (several providers are in the USA; that needs a legal basis for international transfers, which the lawyer in task 3 checks).

2.4 AFTERWARDS

5. Send me: "DPA signed with Supabase / Google / Deepgram: yes or no, and where the audio is processed". I put the right names into the privacy text and the list of service providers for the lawyer.

UNTIL THIS IS DONE

Use only invented data (like in the Test Plan) and your own voice. Do not record real customers' names or addresses.

3. Lawyer review

Why: privacy text, terms and working-time questions are legal texts, not software **Time:** weeks (appointment first)

Look for a lawyer who handles **IT law and data protection** (Fachanwalt für IT-Recht), ideally someone who has done software start-ups. Ask for a **fixed price** for the package below before you start.

3.1 WHAT TO BRING

- **Zunft_Guide.pdf** (what the app does) and the privacy screen of the app: Betrieb → Datenschutz (this is the draft employee notice).
- The list of service providers: Supabase (EU), Deepgram, Google Gemini (see task 2) and what each receives.
- A list of what is stored about people: employees (name, e-mail, role, assignments, notes with voice, appointments, working time), customers of the craftsmen (name, address, phone, e-mail), receipts (photos), recordings (deleted 90 days after the job closes by default).

3.2 WHAT TO ASK FOR

#	Task for the lawyer
3.1	**Privacy policy** for the app and the website, and the finished **employee notice** (the app shows it before someone joins a business)
3.2	**Contract with your customers** (the craftsmen): terms of use, and the **data-processing agreement (AVV)** in which you are the processor and the craftsman the controller; list of sub-processors
3.3	**International transfers:** is the use of Deepgram and Google (USA) covered (standard contractual clauses, EU-US framework)? Is audio of customers and employees acceptable? Is it enough to delete audio 90 days after a job closes?
3.4	**Liability wording:** the AI only proposes; the craftsman is responsible for the content, VAT rate and amounts of the invoice. How to say that in the terms and in the app
3.5	**Working time (Zeiterfassung):** (a) does the tracked time count as a working-time record in the legal sense, and what must a record contain? (b) **may the boss enter or correct time for a worker**? what must the worker be able to see and object to (the app marks such entries "vom Betrieb eingetragen" and the worker sees them in "Meine Zeiten"; there is no change log yet)? (c) is a works council's consent needed? (d) retention periods for time data
3.6	**Account closing:** an owner cannot delete the account in the app because invoices must be kept; the support process for closing a business, and what is deleted when
3.7	**Impressum** and price information rules for the website and the stores
3.8	**Customer data of the craftsmen's customers** in the calendar and the maps link ("Route" opens an address in Google Maps), and the confirmation texts

3.3 AFTERWARDS

1. Send me the finished privacy texts. I put them into the app (replacing the draft) and into the store listing material.
2. Tell me any rule that changes behaviour (for example "the worker must confirm entries" or "keep time data N years"). I build it and test it.

4. Supabase for real use

Why: the free plan pauses a project after a week without use | **Cost:** monthly plan | **Time:** 1-2 hours

> **EVIDENCE**
> Two of your older projects ("UniHub" and "unihub") show the status **INACTIVE**: they were paused. The same would happen to Zunft, and an app whose database is asleep does not work.

4.1 PAID PLAN

1. Dashboard → project **zunft** → **Settings** → **Billing** → **Change plan** → **Pro**. Enter a payment method.
2. Keep the **Spend cap** switched on (so a bug cannot create a huge bill). Open **Database** → **Backups** and check that daily backups are listed.

4.2 THE CONFIRMATION LINK (CURRENTLY LEADS TO AN ERROR PAGE)

When someone registers, Supabase e-mails a link. Today that link points to a page that does not exist. The simplest fix is a tiny web page that says "E-mail confirmed, please open the Zunft app."

3. On `github.com` create a new public repository called `zunft-site`. Add a file `index.html` containing a short sentence (for example: "Danke, deine E-Mail-Adresse ist bestätigt. Du kannst jetzt die Zunft-App öffnen und dich anmelden."). Commit.
4. In the repository: **Settings** → **Pages** → Source: **Deploy from a branch**, branch **main**, folder **/ (root)** → Save. After a minute GitHub shows the address (like `https://rb3111-ship.github.io/zunft-site/`).
5. Supabase → **Authentication** → **URL Configuration**: set **Site URL** to that address and add it under **Redirect URLs**. Save.
6. Authentication → **Sign In / Providers** → **Email**: switch **Confirm email** back **on** (you switched it off for testing). Register a test account and click the link: you must land on your page.

4.3 E-MAIL SENDER

Supabase's built-in e-mail service sends only a handful of mails per hour and is for testing. For real users you need your own sender (SMTP).

7. Create an account at an e-mail sending service (Resend, Brevo or Mailgun have free starter plans). Verify your domain there (task 8 gives you the domain; they show a few DNS entries to copy into your domain settings).
8. Supabase → **Authentication** → **Emails** → **SMTP Settings** → switch on custom SMTP and paste host, port, user, password and a sender address such as `noreply@yourdomain`. Also edit the German texts of the confirmation and password-reset e-mails (Authentication → Emails → Templates).

4.4 AFTERWARDS

9. Tell me when the plan is upgraded and the Site URL is set. Nothing in the app changes. I will then re-check that registration, the confirmation link and the password reset work from a phone.

5. Signing key and Google Play

Why: today the APK is signed with a debug key, which Google Play does not accept **Cost:** one-time Google fee

Time: 1 hour + days of waiting

CRITICAL: THE KEY IS IRREPLACEABLE

The upload key identifies your app. If you lose it or its passwords, updating the app becomes a long support case. Make two copies in different places (for example an encrypted USB stick and a password manager). **Never** put it in Git or send it by e-mail.

5.1 CREATE THE KEY

1. Open a terminal (PowerShell) and run (all on one line; it asks for two passwords and some questions about your name; use real values):

```
D:\Android\jdk17\bin\keytool.exe -genkey -v -keystore D:\secrets\zunft-upload.jks -keyalg RSA
-keysize 2048 -validity 10000 -alias zunft-upload
```

   Create the folder `D:\secrets` first. Write down: the file path, the store password, the key password, and the alias `zunft-upload` in your password manager.

2. Tell me: "key created at D:\secrets\zunft-upload.jks". **Do not send me the passwords in the chat.** I will prepare the build so that it reads the passwords from a local file (`android/key.properties` , excluded from Git) which you create yourself with four lines (storePassword, keyPassword, keyAlias, storeFile). I give you the exact content when you tell me.

5.2 GOOGLE PLAY CONSOLE

3. `play.google.com/console` → create a developer account (one-time fee, identity verification can take several days). Choose **Organisation** if you have a registered business: new *personal* accounts currently have to run a closed test with at least 12 testers for 14 days before they may publish (check Google's current rule).

4. Create the app: name **Zunft**, language German, free app, accept the declarations.

5. Prepare these inputs (I can draft the texts): short description (80 characters), full description, 4-8 phone screenshots, icon 512x512, a **privacy policy URL** (from the lawyer, task 3), and the **Data safety** form. Facts for the Data safety form: the app collects name, e-mail, audio recordings, photos and files, appointments and working time; data is encrypted in transit; users can request deletion; data is shared with processors (Supabase, Deepgram, Google).

6. Testing track: **Testing** → **Internal testing** → create a release → upload the app bundle (I build it with `flutter build appbundle` once the key exists) → add the testers' Gmail addresses → send them the opt-in link. This is the proper way to give the app to pilot craftsmen: updates arrive automatically.

6. Firebase for push messages (done)

STATUS 11 OCTOBER 2026

Your Firebase project `zunft-7118a` is connected. The server part is deployed and tested live (a test message reached Google and was correctly refused for a made-up phone address), and the app part is built. What remains for you is 6.4. The steps below stay as a record of how the project was set up.

Why: so a worker is told when the boss gives him or changes an appointment while his app is closed **Cost:** free

Time: 30 minutes

Today reminders work without Firebase (the phone plans them itself). What is missing is a message *from the boss's action to the worker's phone*. That needs Google's push service (Firebase Cloud Messaging, FCM). You create the project; I build everything else.

6.1 CREATE THE PROJECT

1. Open `console.firebase.google.com` and sign in with your Google account. Click **Create a project** (or **Add project**), name it **Zunft**. Google Analytics: switch it **off** (not needed). Click **Create project**, then **Continue**.
2. On the project page click the **Android** icon ("Add app"). **Android package name**: exactly `de.zunft.zunft_app`. App nickname: Zunft. Leave the SHA-1 field empty for now. Click **Register app**.
3. Click **Download google-services.json**. Save it as `D:\Project_Alpha_X\zunft_app\android\app\google-services.json`. Then click **Next** / **Continue** through the remaining wizard screens (the code changes it shows are my job) and finish with **Continue to console**.

6.2 THE KEY THE SERVER NEEDS

4. Click the gear icon next to "Project Overview" → **Project settings** → tab **Service accounts**. Click **Generate new private key** → **Generate key**. A JSON file downloads.
5. Move it to `D:\secrets\zunft-fcm.json`. This file is a **secret** like a password: never in Git, never in a chat.
6. In **Project settings** → tab **Cloud Messaging** check that **Firebase Cloud Messaging API (V1)** says **Enabled**. If it says Disabled, use the three-dot menu there and open it in Google Cloud to enable it.

6.3 WHAT I DID WITH IT

7. Done: the key is stored as a server secret (the file stays in `D:\secrets`), the app has the Firebase plugin, each phone registers its address, a database trigger tells the server function whenever an appointment is created, moved, cancelled, reassigned or deleted, and a settings switch "Meldungen bei Terminen" turns it on per phone. Test **N45** in the Test Plan checks it on two phones.

6.4 WHAT IS LEFT FOR YOU

8. **Back up the two Firebase files** with your other secrets (task 12): `D:\secrets\zunft-7118a-firebase-adminsdk-...json` (the server key; never put it in Git or send it) and `google-services.json` (a copy is in `D:\secrets` too). The latter is deliberately not in Git; the app cannot be built on another computer without it.

9 Install the new APK on two phones and run **N45**. Each person must switch "Meldungen bei Terminen" on once (Betrieb tab) and allow notifications. Tell me if a notification is late, shows odd text, or does not come at all; that is where phone battery-saver settings sometimes interfere (on some brands you must allow the app to run in the background).

IPHONE

Push on iOS additionally needs an Apple Developer account and a Mac (task 11): in Firebase you would upload an APNs key. Android works without that. Firebase itself costs nothing for this use.

7. Prices, plans and payments

Why: seats and storage are set by hand in SQL today **Time:** a day to decide, a day to set up

> **START FROM THE PROPOSAL**
>
> The document **Zunft_Preise** researches what competitors charge and proposes Solo 14.90 €, Team (5 people) 39 € and Betrieb (10 people) 69 € per month net, yearly = 10 months, 14 days free trial, 9.90 € pilot price. Read it, change the numbers you disagree with, then fill in the table below.

7.1 DECIDE FIRST (WRITE THE ANSWERS DOWN)

Question	Your answer
Plans: for example **Solo** (1 person) and **Team** (up to 5 / 10 people)?	
Monthly price per plan, and extra price per additional person?	
Storage included (default today: 1 GB per business) and price for more?	
Free trial (how many days)? What happens after it?	
What does it cost you per active business (AI, storage, support)? Measure during the pilot before fixing the price.	

7.2 UNTIL PAYMENTS EXIST: SET LIMITS BY HAND

In Supabase → **SQL Editor** → **New query**, replace the business name and run:

```
select id, name, seat_limit, storage_limit_mb from businesses;
update businesses set seat_limit = 5, storage_limit_mb = 5120 where name = 'Elektro Testmann GmbH';
```

The first line lists businesses with their limits; the second changes the seats (people) and the storage in MB for one business. The app cannot change these values, only you can.

7.2B HANDLING SUBSCRIPTIONS BY HAND (DURING THE PILOT)

The app already enforces the subscription; until Stripe is connected **you** are the one who switches a business on and off. Plan about 2 minutes per customer. Everything happens in Supabase → **SQL Editor** → **New query**; paste the text, replace the name in quotes, press **Run**.

> **WHAT THE FOUR STATES MEAN**
>
> `trialing` = free trial (14 days from sign-up, `trial_ends_at`). `active` = paid; `paid_until` is the last paid day (empty = no end, for friends and pilot customers). `past_due` = a payment is late: everything still works 7 days after `paid_until`. `canceled` = works until `paid_until`, then read-only. When access ends the business can still look at everything and export, but cannot create anything new.

1. **See who is who.** Run this to list every business with its state, the owner's e-mail and when it ends:

```
select b.name, u.email as owner_email, b.subscription_status, b.trial_ends_at::date as trial_
ends,
       b.paid_until::date as paid_until, b.seat_limit, b.storage_limit_mb
from businesses b
join profiles p on p.business_id = b.id and p.role = 'owner'
join auth.users u on u.id = p.id
order by b.created_at;
```

2. **A craftsman wants to subscribe.** Send them the payment: a Stripe payment link for their plan (Stripe → Payment links → one link per plan and per month or year) or, for the very first customers, a normal invoice with your bank details. Write down the plan, the amount, the date and the business name.

3. **The money has arrived.** Switch the business on until the end of the period they paid for. Monthly payment: one month ahead; yearly: one year ahead. Pick the plan's seat count (Solo 1, Team 5, Betrieb 10) and keep storage at 1024 MB unless you promised more:

```
update businesses
set subscription_status = 'active',
    paid_until = '2026-11-14',    -- last paid day
    seat_limit = 1,               -- Solo 1, Team 5, Betrieb 10
    storage_limit_mb = 1024
where name = 'Elektro Testmann Gmbh';
```

   Run the list from step 1 again: the row must show `active` and the right date. The person only needs to pull down to refresh in the app; the red strip disappears.

4. **Renewals.** Once a week (for example every Monday) run the list from step 1 and look at `paid_until`. For everyone ending in the next 7 days, send a friendly reminder with the payment link. When the next payment arrives, repeat step 3 with the new date (add one month or one year to the old `paid_until`, not to today, so nobody loses days).

5. **A payment is late.** Do nothing for the first days: after `paid_until` the app shows an amber strip and everything still works for 7 days. To make that visible earlier you can set `subscription_status = 'past_due'`. If nothing arrives after the 7 days the business becomes read-only by itself; contact them politely. Nothing is deleted.

6. **Cancelling.** If a customer cancels, keep their last paid date and set the state so that nobody renews by accident:

```
update businesses set subscription_status = 'canceled' where name = 'Elektro Testmann Gmbh';
```

   It keeps working until `paid_until`, then becomes read-only. They can still export all data. The data stays (invoices must be kept by law); deleting a business is a separate support job, ask me.

7 **Special cases.**

```
-- free for good (pilot customer, friend)
update businesses set subscription_status = 'active', paid_until = null where name = '...';
-- extend a trial by 14 days
update businesses set subscription_status = 'trialing', trial_ends_at = now() + interval '14
days' where name = '...';
-- more people than the plan (a Team that needs 7 seats)
update businesses set seat_limit = 7 where name = '...';
-- test the lock on your own test business: trial ended yesterday
update businesses set subscription_status = 'trialing', trial_ends_at = now() - interval '1 d
ay' where name = 'Elektro Testmann Gmbh';
```

After a lock test set the test business back with the "free for good" line.

CHECK TWICE

The business name must match exactly (capital letters, spaces); if the result says *0 rows*, nothing changed. There are two businesses called Elektro Testmann today (one with a lowercase "t"): look at the list from step 1 first. Never run an `update` without the `where` line: it would change every business.

7.2C THE RENEW BUTTON

The owner sees "Abo verlängern" in the red strip. It opens the page you set at build time with `--dart-define=BILLING_URL=https://...` (your website's subscription page, later a Stripe link). Until you have one, the button says that the payment page is not set up yet. Tell me the address once it exists and I rebuild.

7.3 SET UP STRIPE (WHEN THE PRICES ARE DECIDED)

1 `stripe.com` → create an account with your business details and bank account for payouts. Switch on SEPA direct debit and card payments. Ask your tax advisor about VAT on your own invoices (Stripe Tax can help).

2 **Product catalog** → create one product per plan with a monthly price. Note the price IDs.

3 Tell me the plan names, limits and price IDs. I build: a "Plan" screen, Stripe Checkout and the customer portal through a small server function, and a webhook that sets `seat_limit` and `storage_limit_mb` automatically. You give me the Stripe **test** keys first; the live keys only at launch, set as server secrets by me from a file you create.

7.4 BEFORE THE PUBLIC GOOGLE PLAY RELEASE: CHECK THE RENEW BUTTON

OPEN QUESTION, NOT URGENT FOR THE PILOT

Google Play restricts apps from steering users to outside payments to unlock paid features; the rules differ by region and have been changing. The "Abo verlängern" button opens a payment page on our own website, so it may be a problem once the app is public on Google Play. For the pilot (APK sent directly, or the Play *internal test* track) it does not matter.

1 Before you publish publicly: in the Play Console, open **Policy** → **Payments** (or search "Google Play Payments policy") and read what applies to apps that sell subscriptions to businesses outside the Play store, in Germany and the EU.

2. Decide: (a) the policy allows the link: nothing to do; (b) it does not: tell me, and I change the button to a plain text such as "Verlängere dein Abo auf unserer Website" without a link. People then subscribe and renew on the website only; the app only shows the state.
3. Whichever you choose, keep the prices **and** the pay link off the app screens if the policy asks for it, and tell your tax advisor how you sell (question W8).

8. Domain, e-mail addresses and Impressum

Why: the stores, the privacy text and the e-mail sender all need a real address **Time:** 1-2 hours

1. Check that the name is free: search the German trademark register (`register.dpma.de`) and the web for "Zunft" in software and craft services; check with a lawyer if in doubt. Then register a domain (for example zunft.app or zunft-handwerk.de) at a registrar of your choice.
2. Set up addresses: `support@` (the privacy text and the owner-account closing process refer to a support contact), `datenschutz@` and `noreply@` (for the sender in task 4).
3. Put up a simple website with an **Impressum**, the **privacy policy** and a short product description. The lawyer (task 3) provides the texts. The URLs go into the Play Console listing and into the app.
4. Send me the domain and the support address. I put them into the app text where it says "support" and into the documents.

9. Cost alarms and key hygiene

Why: a bug or an abusive user must not be able to create a large bill **Time:** 30 minutes

1. **Google (Gemini):** budget and alerts as in task 2.2, step 3.
2. **Deepgram:** console → **Billing** → set a usage alert or a spending limit.
3. **Supabase:** keep the spend cap on (task 4.1).
4. Check usage **once a week** during the pilot: Google AI Studio (usage), Deepgram (usage), Supabase (Reports). Write the numbers down; they decide your prices.
5. **Where the secrets are:** the Deepgram, Gemini and cron secrets are stored only as Supabase function secrets; the app contains only the public project key. If you suspect a leak, create a new key at the provider and tell me; I set it with `npx supabase secrets set NAME=value` and redeploy. Never paste a secret key into a chat or a document.

10. The pilot with real craftsmen

Why: only real use shows what is missing **Time:** 2-4 weeks

NOT BEFORE

Tasks 2 (data protection basics), 3.1/3.2 (privacy text and processing agreement at least as a draft), 4 (paid Supabase plan, confirmation link) and 9 (cost alarms). Real customers' data must not go through the app earlier.

1. Pick 2-3 craftsmen you trust: one working alone, one with a helper, ideally one who bills companies (property managers).
2. Give them a written agreement (the lawyer's draft): testing phase, no guarantee, data processing agreement, how to end it and get their data exported.
3. Install through the Play Console internal test track (task 5), or send the APK to the first one in person and sit with them for the first hour.
4. Ask them for one thing weekly: "Which invoice did you correct by hand or not send, and why?" Collect screenshots of every message that confused them.
5. Send me the collected findings. I turn them into fixes, in this order: wrong amounts, lost data, confusing messages, wishes.

11. iPhone version (optional)

IS THE APPLE DEVELOPER PROGRAMME FREE? NO.

- **Apple Developer Program: about 99 USD per year** (a company or an individual). Without it you cannot use TestFlight, cannot publish in the App Store and cannot use push messages.
- A *free* Apple ID lets a developer install an app on *his own* iPhone from a Mac, but it expires after 7 days and cannot be shared with testers. That is only for tinkering.
- You also need a **Mac** (or a paid cloud Mac / build service; some have a small free allowance).
- Registering as a **company** needs a free D-U-N-S number and a website; this takes days to weeks. The fee is waived only for non-profits, schools and government bodies.
- For comparison, Google Play is a **one-time** fee (about 25 USD).

Why: many craftsmen use iPhones **Cost:** Apple fee per year **Needs:** a Mac

iPhone apps can only be built on a Mac (or a cloud Mac service). The code is already written for both systems; what is missing is the build and Apple's paperwork.

1. Join the **Apple Developer Program** at `developer.apple.com/programs` (identity check; organisations need a D-U-N-S number, which takes extra time).
2. Get Mac access: your own Mac, a rented cloud Mac (for example MacinCloud) or a cloud build service for Flutter (for example Codemagic). Ask me which one fits; I prepare the project for it.
3. In **App Store Connect** create the app with the bundle ID I give you; upload via TestFlight to test with a few iPhones.
4. For push on iPhone: create an APNs key (Certificates, Identifiers & Profiles → Keys), upload it in Firebase → Project settings → Cloud Messaging → Apple app configuration (needs task 6).
5. Expect to answer Apple's review questions about privacy (same facts as the Play Data safety form) and to provide a demo account.

12. Name check and backups of your own

Time: 1 hour

1. **Code:** it is on GitHub (`Rb3111-ship/Zunft`). Check on `github.com` that the latest commit is there. Use a strong password and two-factor sign-in on that account.
2. **Keystore and secrets:** two copies of `zunft-upload.jks` and of `zunft-fcm.json` , plus the passwords, in different places (task 5).
3. **Database:** with the Pro plan Supabase makes daily backups (task 4.1). Once a quarter, ask me to test a restore on a scratch project so that you know it works.
4. **These documents:** keep the three PDFs and the signed contracts in one folder (`Zunft-Vertraege` for contracts).
5. **Accounts:** note which e-mail owns Supabase, Google Cloud / Firebase, Deepgram, GitHub, Stripe, Apple and Google Play, so that you are never locked out. Use one owner e-mail you control, plus a recovery option.

13. Languages: have the translations checked

Why: I wrote the translations myself; a native speaker will find wrong or odd wording

Time: 1-2 hours per language

1. Install the app, open **Betrieb → Sprache** (or the language button on the login screen) and choose the language. Walk through the tabs, a job, a new customer, the team screen and the settings.
2. Ask a native speaker, ideally a tradesperson, to read every screen. Let them note wrong, stiff or confusing words with a screenshot. The worst mistakes are in the legal-sounding texts (deleting an account, cancelling an invoice, leaving a business).
3. Send me the list. I change the wording in one place (the translation table) and rebuild. A language can also be removed from the list if nobody checks it.
4. **Lawyer:** the privacy notice and the data-processing terms are still German only. Ask the lawyer (task 3) whether translated versions are needed and who may sign off on them. Until then people who do not read German should have the notice explained by their boss.

> **WHAT STAYS GERMAN ON PURPOSE**
>
> Quotes, invoices, e-invoices, mails and messages to your customers, and the file names are always German. A foreign-language screen never changes what a customer receives.

14. AI settings: the key, the quota and where the audio goes

Why: spoken notes in other languages send the audio to Google; the AI key may be on a free tier **Time:** 30 min

WHAT I SAW

While testing, Google refused some requests of our AI key with "quota exceeded" for the stronger models, and said an older model is not available to new users. That usually means the key is on the **free tier**. On the free tier Google may use the data for improving its products and the limits are small: not suitable for customers' real data or for more than a few users.

1. In `aistudio.google.com` → **API keys** open the key we use and check which **plan** it has. Switch it to a billing account (**Set up billing**), which is the paid tier.
2. Set a budget alarm in Google Cloud Billing (task 9), for example 20 € per month.
3. Read Google's terms for the paid Gemini API: with billing the prompts are not used to train the models, and the processing contract (task 2) applies. Write the answer in your data-protection file.
4. Tell the lawyer (task 3): for German notes the **audio** goes to Deepgram, for notes in other languages the audio goes to **Google**. Both must be in the privacy text and the processing agreements.
5. Tell me when billing is on. I then re-run the language test with the stronger model (for example `gemini-3.8-flash`), which may improve Albanian, Bulgarian and Persian names; today's test used the cheapest model.

Sending me the results

For each task, send a short message like: "Task 4 done: Pro plan, Site URL set to ..., e-mail sender noreply@..., confirmation link works." Where the task says "do not send passwords", send only paths and names. I update the three documents and the working notes after each finished task.